The Islamia University of Bahawalpur

Final Year Project Proposal

AI-Based Research Paper Analysis and Research Gap Identification

Project Title: ResearchPilot: AI Research Assistant

Nature of Project: ☑ Development ☐ Research

Area of Specialization:

☑ Artificial Intelligence (AI) ☐ Machine Learning ☐ Game Development ☐ Internet of Things (IoT)
☐ Data Science and Analytics ☑ Web Development ☐ Cybersecurity ☐ Mobile App Development
☑ Natural Language Processing (NLP) ☐ Blockchain ☐ Image Processing ☐ Other: _____

Introduction:

Students and researchers often need to read many academic papers before they understand what has already been studied about a topic. They must identify the main ideas, compare methods and results, and note the limitations of each paper. Doing this manually can take a lot of time, especially when several papers cover the same subject.

ResearchPilot is a web-based system that will help users find academic articles and analyze research papers in one place. Users will be able to search for publications, upload PDF papers, view structured summaries, ask questions about the documents, and compare information from selected papers. The system will also suggest possible research gaps based on the papers being analyzed and show the evidence used for each suggestion.

The system will use Next.js for the web application, Appwrite for user accounts and data storage, an academic search API such as OpenAlex or Semantic Scholar, and an AI model API for document analysis. The project will focus on making research work easier to organize and verify. The suggestions produced by the system will need to be checked by the user and will not be treated as proof that a topic has never been researched.

Problem Statement:

Finding relevant papers is only one part of a literature review. Students also have to understand each paper, compare its methods and findings with other studies, and identify limitations or questions that need further research. These tasks are often completed manually across different websites, PDF files, and notes, which makes the process slow and difficult to organize.

General AI tools can summarize documents, but their answers may not always be supported by the original papers. It can also be difficult to compare several studies and trace a conclusion back to the source passage. ResearchPilot will address this problem by bringing article search, PDF analysis, evidence-based question answering, article comparison, and potential research gap suggestions into one application.

Project Goal / Objectives:

Goal: To develop a web-based AI research assistant that helps users find, understand, and compare academic papers and identify possible research gaps from the available literature.

1. To provide academic article search through a supported API such as OpenAlex or Semantic Scholar.
2. To allow users to upload PDF papers and extract their text for analysis.
3. To generate summaries covering the purpose, method, findings, and limitations of a paper.
4. To answer questions using relevant passages from the selected papers and provide source references where available.
5. To compare multiple papers by their methods, findings, and stated limitations.
6. To suggest possible research gaps and research questions based on the analyzed papers, with supporting evidence.
7. To save users' papers, projects, and analysis results, and test the main functions of the completed system.

Application Architecture / Technology Stack / Tools / Hardware:

Language(s): TypeScript / JavaScript for the web application. Python may be considered if a separate document-processing or machine-learning component is needed.

Architecture: A web-based client-server architecture. The user interface will be built in Next.js. Appwrite will manage authentication, database records, and file storage. The application will connect to an academic search API and an AI model API. PDF text will be processed and relevant passages will be retrieved to support answers and comparisons.

Libraries / Frameworks: Next.js, Appwrite SDK, a PDF text extraction library, an embedding model, and a suitable retrieval or vector-search component. The final libraries will be selected during implementation.

AI Method: Large Language Model API and Retrieval-Augmented Generation (RAG) for document-based answers, summaries, comparisons, and potential research gap suggestions.

Academic Search: OpenAlex API or Semantic Scholar API. The final choice will depend on API access, available metadata, usage limits, and project requirements. The project will not depend on an assumed official public Google Scholar API.

Tools: Visual Studio Code, Git and GitHub, a web browser, and suitable testing tools.

Hardware: A standard laptop or desktop computer with internet access. A dedicated GPU is not expected to be required because the project will use an existing AI model through an API.

Expected Outcome:

The expected outcome is a working web application where users can search for academic papers, upload PDFs, generate summaries, ask questions with source references, compare selected papers, and review possible research gaps. The system will be tested using a selected set of papers. Its results will be checked for relevance and support from the source documents.

Student Data:

Sr. No.	Registration No.	Name	Phone	Signature
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References (if any):

1. Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems. https://arxiv.org/abs/2005.11401
2. OpenAlex. OpenAlex Documentation. https://docs.openalex.org/
3. Semantic Scholar. Academic Graph API Documentation. https://api.semanticscholar.org/api-docs/
4. Next.js. Official Documentation. https://nextjs.org/docs
5. Appwrite. Official Documentation. https://appwrite.io/docs

Project Supervisor:

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